```
> //CREATE NEW DATABASE
> use studentDB
< switched to db studentDB
```

```
> //CREATE A COLLECTION
> db.createCollection("students")
< { ok: 1 }
```

```
> //INSERT ONE
> db.students.insertOne({name:"chithra",age:"25",course:"MERN stack",status:"En rolled"})
< {
    acknowledged: true,
    insertedId: ObjectId('6ae1c9aba3c6d6a1ae6c447f')
  }
```

```
> /* INSERTMANY */
> db.students.insertMany([{name:"Aravind",age:31,course:"Gen AI",status:"En rolled"},
                          {name:"Jeya",age:28,course:"Python",status:"Completed"},
                          {name:"vijay",age:19,course:"MERN stack",status:"In progress"},
                          {name:"surya",age:24,course:"Java full stack",status:"completed"},
                          {name:"meenu",age:32,course:"MERN stack",status:"In progress"},
                          {name:"mathu",age:22,course:"Python",status:"En rolled"}])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ae1b1d7a3c6d6a1ae6c4478'),
      '1': ObjectId('6ae1b1d7a3c6d6a1ae6c4479'),
      '2': ObjectId('6ae1b1d7a3c6d6a1ae6c447a'),
      '3': ObjectId('6ae1b1d7a3c6d6a1ae6c447b'),
      '4': ObjectId('6ae1b1d7a3c6d6a1ae6c447c'),
      '5': ObjectId('6ae1b1d7a3c6d6a1ae6c447d')
    }
  }
```

```
> //FETCH ALL DOCUMENTS
> db.students.find()
< {
    _id: ObjectId('6ae1c9aba3c6d6a1ae6c447f'),
    name: 'chithra',
    age: '25',
    course: 'MERN stack',
    status: 'En rolled'
  }
  {
    _id: ObjectId('6ae1ca71a3c6d6a1ae6c4480'),
    name: 'Aravind',
    age: 31,
    course: 'Gen AI',
    status: 'En rolled'
  }
  {
    _id: ObjectId('6ae1ca71a3c6d6a1ae6c4481'),
    name: 'Jeya',
    age: 28,
    course: 'Python',
    status: 'Completed'
```

```
}
{
  _id: ObjectId('6ae1ca71a3c6d6a1ae6c4482'),
  name: 'vijay',
  age: 19,
  course: 'MERN stack',
  status: 'In progress'
}
{
  _id: ObjectId('6ae1ca71a3c6d6a1ae6c4483'),
  name: 'surya',
  age: 24,
  course: 'Java full stack',
  status: 'completed'
}
{
  _id: ObjectId('6ae1ca71a3c6d6a1ae6c4484'),
  name: 'meenu',
  age: 32,
  course: 'MERN stack',
  status: 'In progress'
}
```

```
{
  _id: ObjectId('6ae1b1d7a3c6d6a1ae6c447d'),
  name: 'mathu',
  age: 22,
  course: 'Python',
  status: 'En rolled'
}
```

```
> //FETCH MERN STACK COURSE DOCUMENTS
> db.students.find({course:"MERN stack"})
< {
    _id: ObjectId('6ae1c9aba3c6d6a1ae6c447f'),
    name: 'chithra',
    age: '25',
    course: 'MERN stack',
    status: 'En rolled'
  }
  {
    _id: ObjectId('6ae1ca71a3c6d6a1ae6c4482'),
    name: 'vijay',
    age: 19,
    course: 'MERN stack',
    status: 'In progress'
  }
```

```
{
  _id: ObjectId('6ae1ca71a3c6d6a1ae6c4484'),
  name: 'meenu',
  age: 32,
  course: 'MERN stack',
  status: 'In progress'
}
```

```
> //UPDATE SINGLE DOCUMENT
> db.students.updateOne({name:"vijay"},{$set:{status:"completed"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 0,
    upsertedCount: 0
  }
> db.students.find({name:"vijay"})
< {
    _id: ObjectId('6ae1b1d7a3c6d6a1ae6c447a'),
    name: 'vijay',
    age: 19,
    course: 'MERN stack',
    status: 'completed'
  }
```

```
> //UPDATE MANY
> db.students.updateMany({course:"MERN stack"},{$set:{course:"Full stack"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
> db.students.find({course:"Full stack"})
< {
    _id: ObjectId('6ae1c9aba3c6d6a1ae6c447f'),
    name: 'chithra',
    age: '25',
    course: 'Full stack',
    status: 'En rolled'
  }
```

```
{
  _id: ObjectId('6ae1ca71a3c6d6a1ae6c4482'),
  name: 'vijay',
  age: 19,
  course: 'Full stack',
  status: 'completed'
}
{
  _id: ObjectId('6ae1ca71a3c6d6a1ae6c4484'),
  name: 'meenu',
  age: 32,
  course: 'Full stack',
  status: 'In progress'
}
```

```
> //DELETE ONE DOCUMENT
> db.students.deleteOne({course:"Gen AI"})
< {
    acknowledged: true,
    deletedCount: 1
  }
```

```
> //DELETE ALL
> db.students.deleteMany({})
< {
    acknowledged: true,
    deletedCount: 6
  }
> db.students.find()
<
studentDB>
```

```
> //APPLY $gt
> db.students.find({age:{$gt:30}})
< {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c4488'),
    name: 'Aravind',
    age: 31,
    course: 'Gen AI',
    status: 'En rolled'
  }
  {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c448c'),
    name: 'meenu',
    age: 32,
    course: 'MERN stack',
    status: 'In progress'
  }
```

```
> //APPLY $lt
> db.students.find({age:{$lt:20}})
< {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c448a'),
    name: 'vijay',
    age: 19,
    course: 'MERN stack',
    status: 'In progress'
  }
```

```
> //APPLY $in
> db.students.find({status:{$in:["In progress","completed"]}})
< {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c448a'),
    name: 'vijay',
    age: 19,
    course: 'MERN stack',
    status: 'In progress'
  }
  {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c448b'),
    name: 'surya',
    age: 24,
    course: 'Java full stack',
    status: 'completed'
  }
```

```
{
  _id: ObjectId('6ae1d92fa3c6d6a1ae6c448c'),
  name: 'meenu',
  age: 32,
  course: 'MERN stack',
  status: 'In progress'
}
```

```
> //APPLY $and
> db.students.find({$and:[{age:{$gt:30}},{course:"MERN stack"}]})
< {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c448c'),
    name: 'meenu',
    age: 32,
    course: 'MERN stack',
    status: 'In progress'
  }
```

```
> //APPLY $or
> db.students.find({$or:[{course:"Gen AI"},{age:25}]})
< {
    _id: ObjectId('6ae1d925a3c6d6a1ae6c4487'),
    name: 'chithra',
    age: 25,
    course: 'MERN stack',
    status: 'En rolled'
  }
  {
    _id: ObjectId('6ae1d92fa3c6d6a1ae6c4488'),
    name: 'Aravind',
    age: 31,
    course: 'Gen AI',
    status: 'En rolled'
  }
```

```
> db.students.insertOne({name:"adhirai",age:24,email:"adhirai@gmail.com"})
< {
    acknowledged: true,
    insertedId: ObjectId('6ae1de46a3c6d6a1ae6c448e')
  }
> //APPLY $exists
> db.students.find({email:{$exists:true}})
< {
    _id: ObjectId('6ae1de46a3c6d6a1ae6c448e'),
    name: 'adhirai',
    age: 24,
    email: 'adhirai@gmail.com'
  }
```

```
> //CREATE DATABASE
> use E-commerce
< switched to db E-commerce
> //CREATE COLLECTION
> db.createCollection("products")
< { ok: 1 }
> //INSERT
> db.products.insertMany([{name:"T-shirts",price:499,stock:"available"},
                          {name:"Jeans",price:999,stock:"out of stock"},
                          {name:"Shirt",price:699,stock:"available"}])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ae2ec4366d57763fa9b59e3'),
      '1': ObjectId('6ae2ec4366d57763fa9b59e4'),
      '2': ObjectId('6ae2ec4366d57763fa9b59e5')
    }
  }
```

```
> //LIST
> db.products.find()
< {
    _id: ObjectId('6ae2ec4366d57763fa9b59e3'),
    name: 'T-shirts',
    price: 499,
    stock: 'available'
  }
  {
    _id: ObjectId('6ae2ec4366d57763fa9b59e4'),
    name: 'Jeans',
    price: 999,
    stock: 'out of stock'
  }
  {
    _id: ObjectId('6ae2ec4366d57763fa9b59e5'),
    name: 'Shirt',
    price: 699,
    stock: 'available'
  }
```

```
> //SEARCH PRODUCT
> db.products.find({stock:"available"})
< {
    _id: ObjectId('6ae2ec4366d57763fa9b59e3'),
    name: 'T-shirts',
    price: 499,
    stock: 'available'
  }
  {
    _id: ObjectId('6ae2ec4366d57763fa9b59e5'),
    name: 'Shirt',
    price: 699,
    stock: 'available'
  }
```

```
> //UPDATE PRODUCT
> db.products.updateOne({name:"Jeans"},{$set:{stock:"sold out"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.find({name:"Jeans"})
< {
    _id: ObjectId('6ae2ec4366d57763fa9b59e4'),
    name: 'Jeans',
    price: 999,
    stock: 'sold out'
  }
```

```
> //DELETE PRODUCT
> db.products.deleteOne({name:"T-shirts"})
< {
    acknowledged: true,
    deletedCount: 1
  }
> db.products.find()
< {
    _id: ObjectId('6ae2ec4366d57763fa9b59e4'),
    name: 'Jeans',
    price: 999,
    stock: 'sold out'
  }
  {
    _id: ObjectId('6ae2ec4366d57763fa9b59e5'),
    name: 'Shirt',
    price: 699,
    stock: 'available'
  }
```